Final Project Report Guidelines

MAT 219, Fall 2026

Your final report is the revised and completed version of your project draft. Present a coherent investigation that connects your research question, data, statistical analysis, and conclusions. Incorporate feedback on your draft and presentation, and write so that a reader can understand the project without seeing your earlier submissions.

Due: Monday, December 14 by 12:00 pm

What to include

1. **Introduction**

   State your specific research question(s) and explain the motivation and relevance of the topic. Reuse and revise appropriate material from your proposal and draft. Include a descriptive project title and all group members' names at the beginning of the report.

2. **Data description and exploratory analysis**

   Identify the data source and provide a working link. Explain what one observation represents, the sample size used, the population or time period covered, and the relevant variables. Describe missing values, exclusions, filtering, merging, or other preparation. Include selected summary statistics and graphs that help readers understand the data and patterns relevant to your question. There is no minimum graph count for the final report; select useful visuals rather than including every graph from your draft.

3. **Statistical methods and results**

   Complete the analysis using appropriate methods from the course. Explain why each method fits your question and variables, identify the population quantity of interest, and check the relevant conditions. For a hypothesis test, state the null and alternative hypotheses. Report the numerical results appropriate to your method, such as estimates, confidence intervals, test statistics, p-values, or regression results, and interpret them in context. Include tables or graphs of the main analysis results where useful. These serve a different purpose from exploratory visuals; you may refer back to earlier figures without repeating them.

4. **Discussion and conclusion**

   Answer your research question using the evidence from your analysis. Discuss the size and practical meaning of findings, not only statistical significance. Explain important limitations, including possible bias, independence concerns, generalizability, and whether causal claims are justified. Identify remaining uncertainties or sensible directions for further investigation. Avoid repeating all numerical results.

5. **References**

   Cite the dataset, outside sources, and R packages used. Use a consistent citation format and provide enough information for readers to locate each source.

Revision note

At the end of the report, add a short paragraph or bulleted list describing the main changes made after draft feedback and presentation questions. Address important concerns about your data, methods, or interpretation. If you did not adopt a suggestion, briefly explain why.

Group report format and submission

- Group submission: Submit one PDF report per group on Canvas by December 14, together with the R code and data or loading instructions described below. List all group members on the report.
- Aim for 1,200-1,800 words, excluding tables, figure captions, code, references, and the revision note. The individual reflection is a separate submission and is not included in this word count.
- Number and label figures and tables, including axes and units where applicable. Refer to each one in the text and explain what readers should learn from it.
- Present selected results clearly rather than pasting lengthy R output. Keep the main report readable and submit the code separately.

R code and data submission

Submit your final .Rmd file with the PDF report. Include data loading, preparation, summary statistics, graphs, and statistical analysis, with brief comments explaining the steps. Provide the data file(s), or a direct download link and clear loading instructions.

Run the complete code from a fresh R session before submitting. It should reproduce the figures, tables, and numerical results in the report. Include any required packages and avoid relying on objects created earlier or file paths that work only on your computer.

How the final report is graded

The report is graded on statistical reasoning, accuracy, clarity, and completion. A statistically significant result is not required. An appropriate analysis with a careful explanation of an inconclusive result is stronger than an unsupported claim.

This will be graded on whether…	✓
The research question is clear and the report can be understood on its own	☐
The data, preparation, and selected exploratory findings are explained	☐
Statistical methods are appropriate and relevant conditions are checked	☐
Results are accurate, reproducible, and interpreted in context	☐
Conclusions are supported and limitations are thoughtfully addressed	☐
Important feedback is addressed and changes are described in the revision note	☐
The report, references, code, and data instructions are complete and organized	☐

Individual reflection submission

Each student must submit a separate 150-200 word reflection on Canvas by December 14. Submit it individually, not as part of the group report. Include your name and project title. Write independently in your own words, without AI-generated text, and address all three prompts below.

1. Your contribution: Describe the specific work you personally completed. Explain any changes from the role allocation plan submitted with your draft.
2. Your statistical understanding: Explain one statistical decision your group made and why it was appropriate, or explain one main numerical result in the context of your research question. Refer to a specific method, result, figure, or table from your project.
3. Your AI use: Identify any AI tools you personally used for this project, explain what you used them for, and describe how you checked their suggestions. If you did not use AI, state that explicitly.

The reflection is assessed individually for specific evidence of contribution, statistical understanding, and a clear account of AI use. It should describe your own experience rather than repeat the group report.

A note on using AI tools

The report must reflect your group's own understanding. If you use AI tools to help with code or explanations, verify the results against your data and course materials. Every group member should be able to explain the methods, code, and conclusions submitted.